

Mini-Project 6.3 Report

Predictive Modeling of Student Dropout Risk Using Supervised Machine Learning Techniques.

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March, 2026

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Introduction

This project applies supervised machine learning techniques to predict student dropout risk. The objective is to develop a predictive framework capable of identifying students at risk early in their academic journey, enabling timely interventions and data-driven decision making. Accurate predictive models can support educational institutions in improving student retention, optimizing resource allocation, and enhancing academic support strategies.

Methodology

This study implemented a dual-architecture machine learning pipeline to analyze and classify student data across three distinct stages. The workflow was designed to ensure technical rigor, from initial data inspection through data preprocessing, feature engineering, model development, hyperparameter optimization, and comparative evaluation (Figure 1).

Results

1) EDA and Data Preprocessing:

The student's information was loaded, inspected and described to understand the data architecture. Initial analysis focused on identifying structural metadata, missing values, and feature distributions.

Data cleaning involved removing columns with extremely high categorical cardinality (>200 unique values) and columns containing more than 50% missing values.

Rows containing missing values less than 2% of the dataset were removed and median imputation was applied for other missing rows to preserve data integrity and avoid unnecessary loss of observations.

Ordinal encoding used to convert features that display hierarchy in their category into numerical values (Table .1). Binary encoding was applied to columns with nominal features while the remaining columns were converted to numerical data using one-hot encoding method.

Feature Engineering was employed to convert date of birth to datetime and consequently create a feature (Age) from date of birth. Features not relevant to the analysis were dropped. CompletedCourse was renamed DropOut to enhance understanding of the target variable (Table.1)

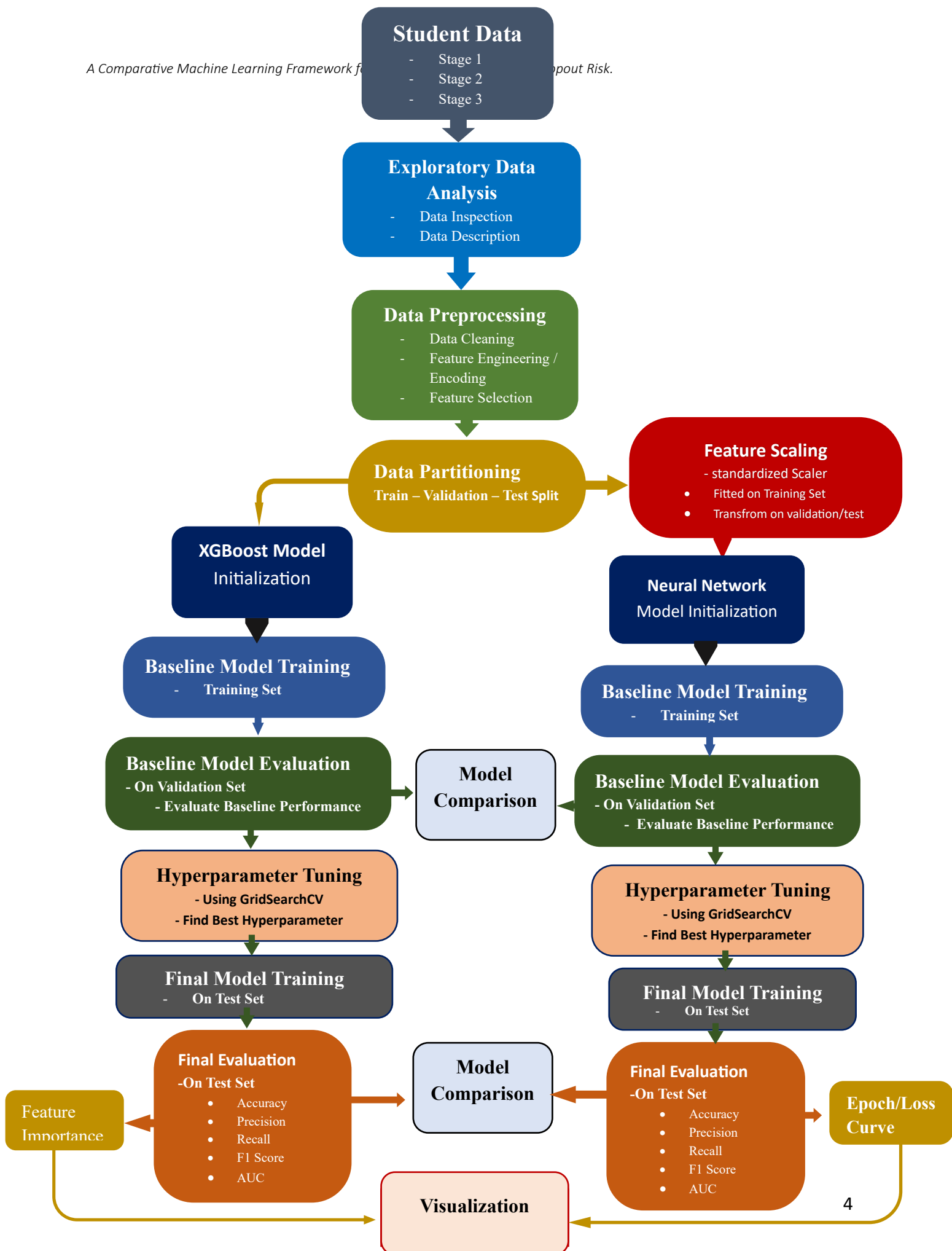


Figure 1: Schematic Overview of Student Data Analysis Pipeline

Table 1: Encoded columns in numerical format

CourseLevel	DropOut	IsFirstIntake	Gender
3	0	1	0
0	0	0	0
2	0	0	0
2	0	1	0
3	0	1	0

2) Data Partitioning:

The preprocessed dataset was separated into:

- **Feature variables (X):** Independent variables used to predict student dropout.
- **Target variable (y):** The dependent variable indicating whether a student dropped out or completed the course.

The data was then split using the **train-test split method**, ensuring reproducibility by setting a random seed, as seen in Table 2.

Table 2: Data partitioning

Dataset	Purpose	Proportion of Data
Training Set	Model learning and pattern recognition	80%
Test Set	Model evaluation and performance assessment	20%
Validation Set	Used to evaluate model performance during tuning.	10% of training set

3) Model Development and Optimization:

To achieve the research objectives, two supervised learning models were employed, as detailed in Table 3:

Table 3: Architecture of Predictive Models

Model	Type	Operation	Key Parameters
XGBoost	Tree-based ensemble model	Gradient Boosting	Learning rate, max depth, number of estimators
Neural Network	Multi-Layer Perceptron (MLP)	Forward Propagation	Number of neurons, activation function, optimizer

a) XGBoost Model:

A baseline XGBoost classifier was initially trained on the training dataset to establish baseline performance. Hyperparameter optimization was subsequently performed using GridSearchCV within Scikit-learn, where multiple parameter combinations were evaluated through cross-validation to identify the optimal configuration (Table 4). The model was then retrained using the best-performing hyperparameters to produce the final tuned model. Model performance was evaluated on the validation/test dataset using several evaluation metrics. The tuned model's performance was compared with the baseline model to assess improvements resulting from hyperparameter optimization. In addition, a comparative analysis was conducted across the three stages of student data, and the performance of the XGBoost model was further compared with that of a Neural Network model.

Table 4: XGboost Hyperparamater grid

Hyperparameters	Low (Level 1)	Medium (Level 2)	High (Level 3)
n_estimator	100	200	300
Max_depth	3	5	7
Learning_rate	0.01	0.1	0.2

b) Neural Network Model and Scaling

Input features were scaled using a StandardScaler technique prior to model training, as Neural Networks are highly sensitive to the magnitude of input features. To preserve the integrity of the evaluation process, the scaler was fitted only on the training dataset and subsequently applied to the validation and test datasets to prevent data leakage.

A baseline Sequential Neural Network model (Table 5) was then instantiated, compiled, and trained on the training dataset to establish initial performance. Subsequently, hyperparameter optimization was conducted using GridSearchCV, where multiple parameter combinations (Table 6) were evaluated through cross-validation to determine the optimal configuration. The same evaluation procedure used for the XGboost model was followed for performance assessment. Training progress was validated using **Epoch/Loss curves** to monitor for overfitting and ensure stable gradient descent (Figure 2).

Table 5: A Simple Baseline Sequential Neural Network Model

Layer(type)	Output Shape	Param #	Description
Dense(Input layer)	(None, 64)	24,896	Input layer
Dense(Hidden layer)	(None, 32)	2,080	Activation function
Dense(Output layer)	(None, 1)	33	Sigmoid output for binary classification

Table 6: A Neural Network Hyperparameter Grid

Hyperparameter	Level 1	Level 2	Level 3
Number of Neurons	32	64	128
Activation Function	ReLU	Tanh	Sigmoid
Optimizer	Adam	SGD	RMSprop

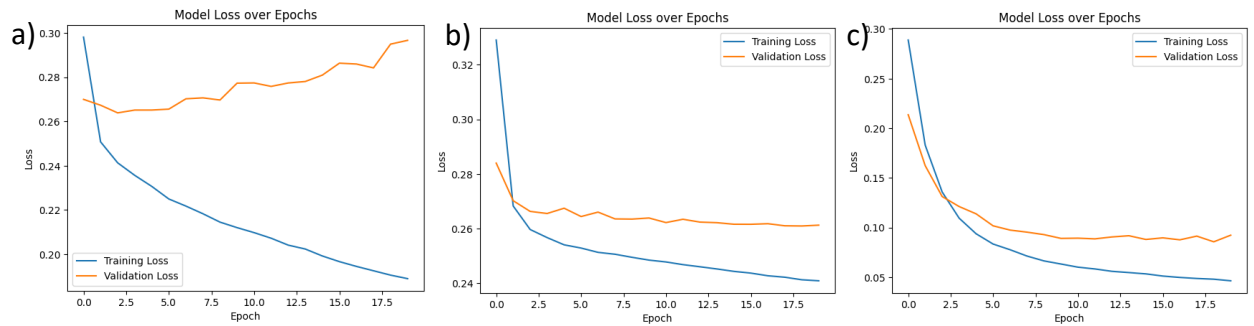


Figure 2: Epoch/Loss Curves across a) Stage 1, b) Stage 2, c) Stage 3

4) Comparative Evaluation:

The final stage utilized a unified suite of metrics to evaluate performance across the three data stages:

a) Stage 1

Table 7: Baseline Model Performance Comparison (Stage 1 Data)

Metric	XGboost Baseline Model (Stage 1)	Neural Network Baseline Model (Stage 1)
Accuracy	0.8911	0.8891
Precision	0.6661	0.6432
Recall	0.5473	0.5832
F1-Score	0.6009	0.6117
ROC AUC Score	0.8845	0.7631

Table 8: Tuned Model Performance Comparison (Stage 1 Data):

Metric	Tuned XGBoost (Stage 1)	Tuned Neural Network (Stage 1)
Accuracy	0.8935	0.8899
Precision	0.6830	0.6707
Recall	0.5393	0.5206
F1-Score	0.6027	0.5862
ROC AUC Score	0.8838	0.7378

b) Stage 2

Table 9: Baseline Model Performance Comparison (Stage 2)

Metric	XGboost Baseline Model (Stage 2)	Neural Network Baseline Model (Stage 2)
Accuracy	0.8974	0.8982
Precision	0.6927	0.6656
Recall	0.5277	0.6011
F1-Score	0.5991	0.6317
ROC AUC Score	0.8991	0.8643

Table 10: Tuned XGBoost vs. Tuned Neural Network Performance Comparison (Stage 2 Data)

Metric	XGBoost Tuned (Stage 2)	Neural Network Tuned (Stage 2)
Accuracy	0.8994	0.8974
Precision	0.7079	0.6563
Recall	0.5235	0.6163
F1-Score	0.6019	0.6357
ROC AUC	0.9002	0.8763

c) Stage 3

Table 11: Baseline Model Performance Comparison (Stage 3)

Metric	XGBoost Model	Neural Network Model
Accuracy	0.9755	0.9650
Precision	0.9286	0.9177
Recall	0.9003	0.8338
F1-Score	0.9142	0.8737
ROC AUC	0.9928	0.9679

Table 11: Tuned Models Performance Comparison - Stage 3

Metric	XGBoost Tuned (Stage 3)	Neural Network Tuned (Stage 3)
Accuracy	0.9761	0.9710
Precision	0.9338	0.9188
Recall	0.8989	0.8781
F1-Score	0.9160	0.8980
ROC AUC	0.9926	0.9854

Interpretation:

Across all three stages, the XGBoost model consistently achieved slightly higher ROC-AUC scores than the Neural Network, indicating stronger discrimination capability in identifying students at risk of dropout. However, the neural network demonstrated competitive performance in terms of recall, suggesting better detection of dropout cases in certain stages.

Conclusion

This research confirms that machine learning is a powerful tool for forecasting student attrition. Among the models tested, the XGBoost classifier consistently achieved superior predictive performance across most evaluation metrics. Notably, the accuracy of these predictions increased as students advanced through the program, indicating that data from later academic stages offers the most reliable indicators of dropout risk. Ultimately, these results demonstrate that implementing predictive analytics can empower institutions to launch targeted early-intervention programs and significantly boost student retention.

Since predictive performance improved significantly across the three stages of student data, institutions should establish systems for **continuous monitoring of student progression**. Periodic analysis of student performance data can help detect emerging dropout risks as students advance through their academic programs.

Incorporating richer datasets may further improve predictive performance and provide deeper insights into the factors contributing to student attrition.